

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: PF	Course Code: CMPE-112L
Assignment Type: lab	Dated: 25 MAY 2026
Semester: 1st	Session: Spring 2026
Lab/Project/Assignment #: 12	CLOs to be covered: CLO2
Lab Title: Lambda functions	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 12

Lab Goals/Objectives:

- To understand anonymous functions.
- To create and use lambda functions.
- To write short and efficient code.
- To use lambda functions with built-in functions

Equipment Required:

- Computer/Laptop
- Python 3.x
- VS Code / PyCharm / IDLE

Theory:

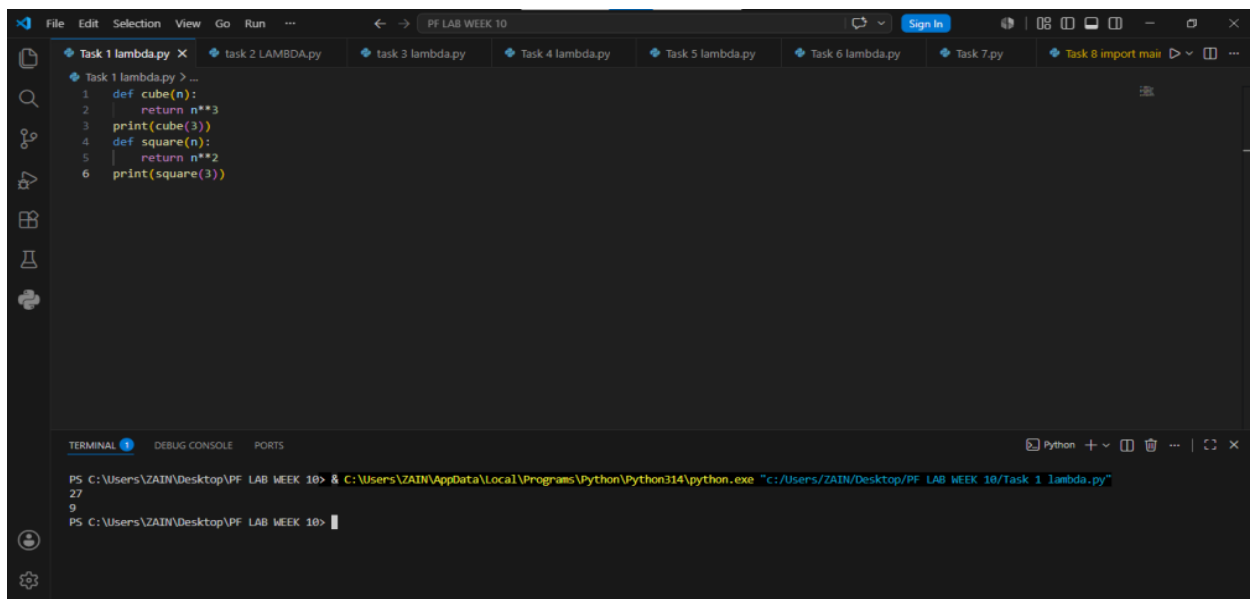
A lambda function is a small anonymous function created using the `lambda` keyword. It can take any number of arguments but contains only one expression. Lambda functions are commonly used for short operations where defining a complete function is unnecessary. They are often used with functions like `map()`, `filter()`, and `sorted()` to write cleaner and more concise code.

Lab Work:

Task 1:

Write the given Python program that uses normal `def` functions to calculate cube and square of a number by using lambda functions instead.

Result:



The screenshot shows a VS Code editor window with a file explorer on the left and a terminal at the bottom. The editor displays a Python file named 'Task 1 lambda.py' with the following code:

```
1 def cube(n):
2     return n**3
3 print(cube(3))
4 def square(n):
5     return n**2
6 print(square(3))
```

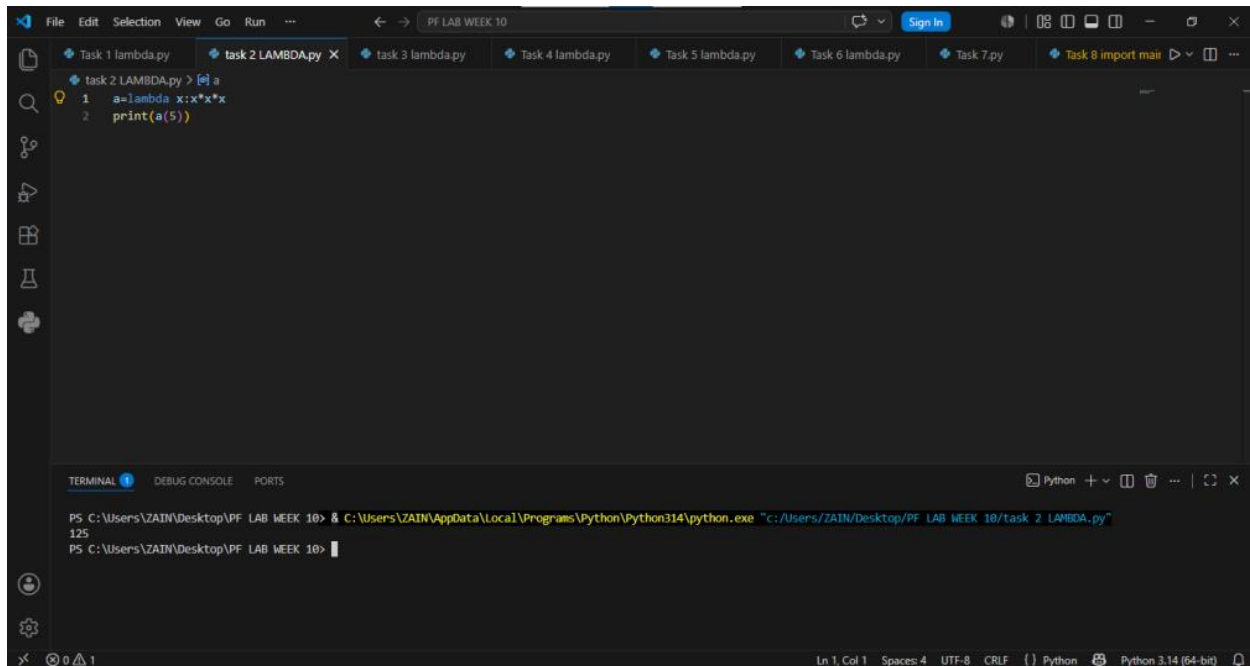
The terminal at the bottom shows the command prompt output:

```
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10> & C:\Users\ZAIN\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/ZAIN/Desktop/PF LAB WEEK 10/Task 1 lambda.py"
27
9
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10>
```

Task 2:

Write a Python program using a lambda function to calculate the cube of a number and print the result.

Result:



The screenshot shows a Python IDE with a file explorer on the left and a terminal at the bottom. The file explorer shows several files, including 'task 2 LAMBDA.py'. The code in 'task 2 LAMBDA.py' is as follows:

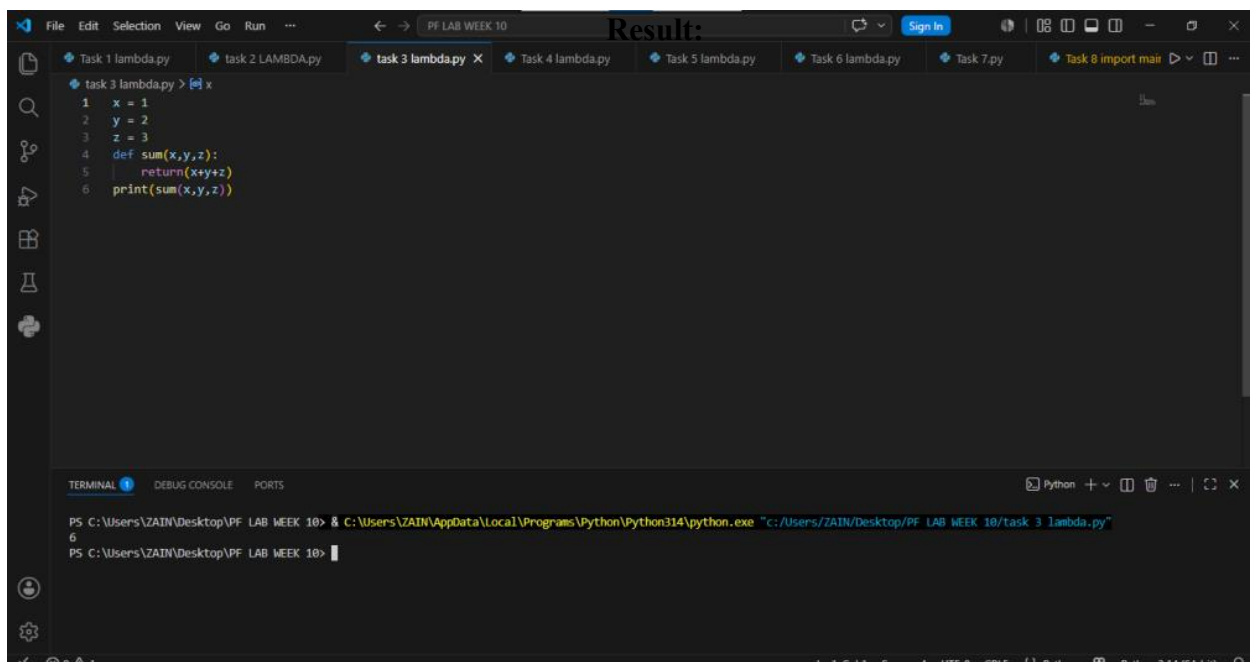
```
1 a=lambda x:x*x*x
2 print(a(5))
```

The terminal shows the command to run the script and the output:

```
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10> & C:\Users\ZAIN\AppData\Local\Programs\Python\Python314\python.exe "C:/Users/ZAIN/Desktop/PF LAB WEEK 10/task 2 LAMBDA.py"
125
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10>
```

Task 3:

Write a Python program that uses a lambda function to calculate the sum of three numbers then print the result.



The screenshot shows a Python IDE with a file explorer on the left and a terminal at the bottom. The file explorer shows several files, including 'task 3 lambda.py'. The code in 'task 3 lambda.py' is as follows:

```
1 x = 1
2 y = 2
3 z = 3
4 def sum(x,y,z):
5     return(x+y+z)
6 print(sum(x,y,z))
```

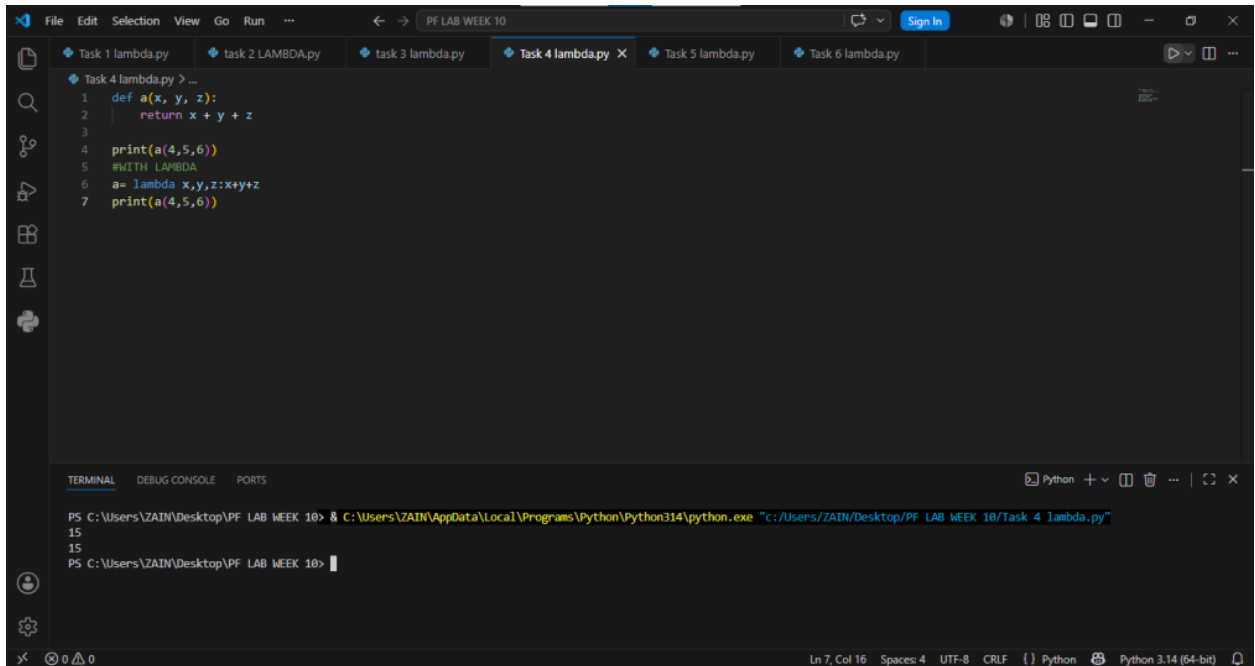
The terminal shows the command to run the script and the output:

```
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10> & C:\Users\ZAIN\AppData\Local\Programs\Python\Python314\python.exe "C:/Users/ZAIN/Desktop/PF LAB WEEK 10/task 3 lambda.py"
6
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10>
```

Task 4:

Write a Python program to create a lambda function that takes three numbers as input and returns their sum

Result:



The screenshot shows a Python IDE with a file named 'Task 4 lambda.py'. The code defines a function 'a(x, y, z)' that returns the sum of three numbers, and then uses a lambda function to calculate the sum of 4, 5, and 6. The terminal output shows the command to run the script and the resulting output '15'.

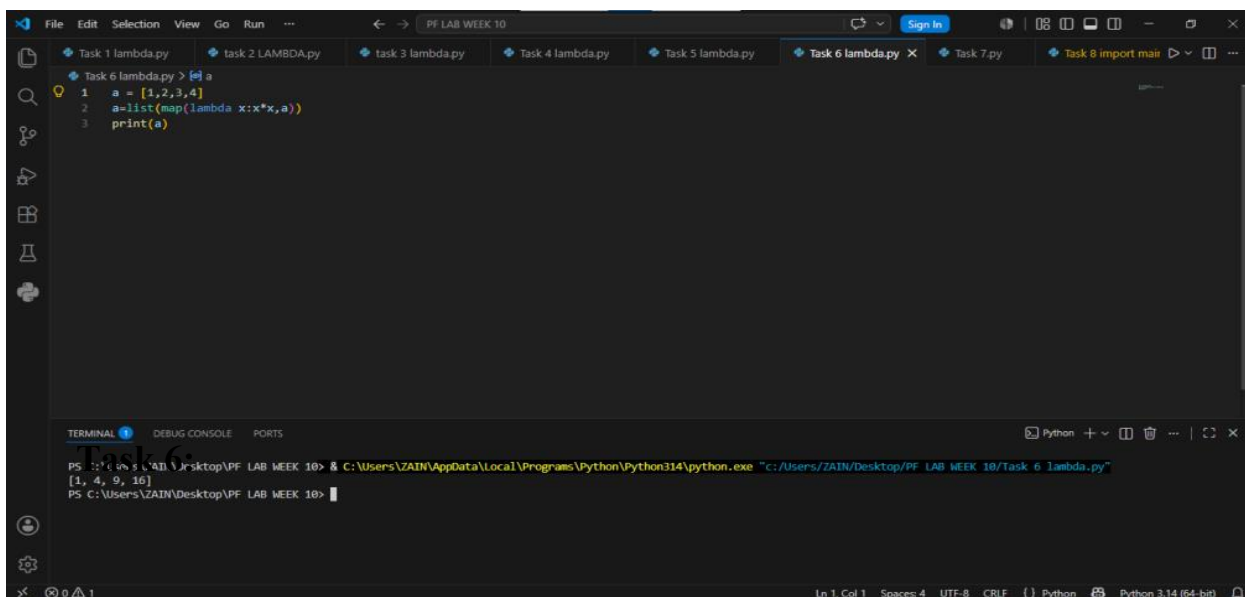
```
1 def a(x, y, z):
2     return x + y + z
3
4 print(a(4,5,6))
5 #WITH LAMBDA
6 a= lambda x,y,z:x+y+z
7 print(a(4,5,6))
```

```
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10> & C:\Users\ZAIN\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/ZAIN/Desktop/PF LAB WEEK 10/Task 4 lambda.py"
15
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10>
```

Task 5:

Write a Python program using a lambda function with to square all elements in the list and print the updated list.

Result:



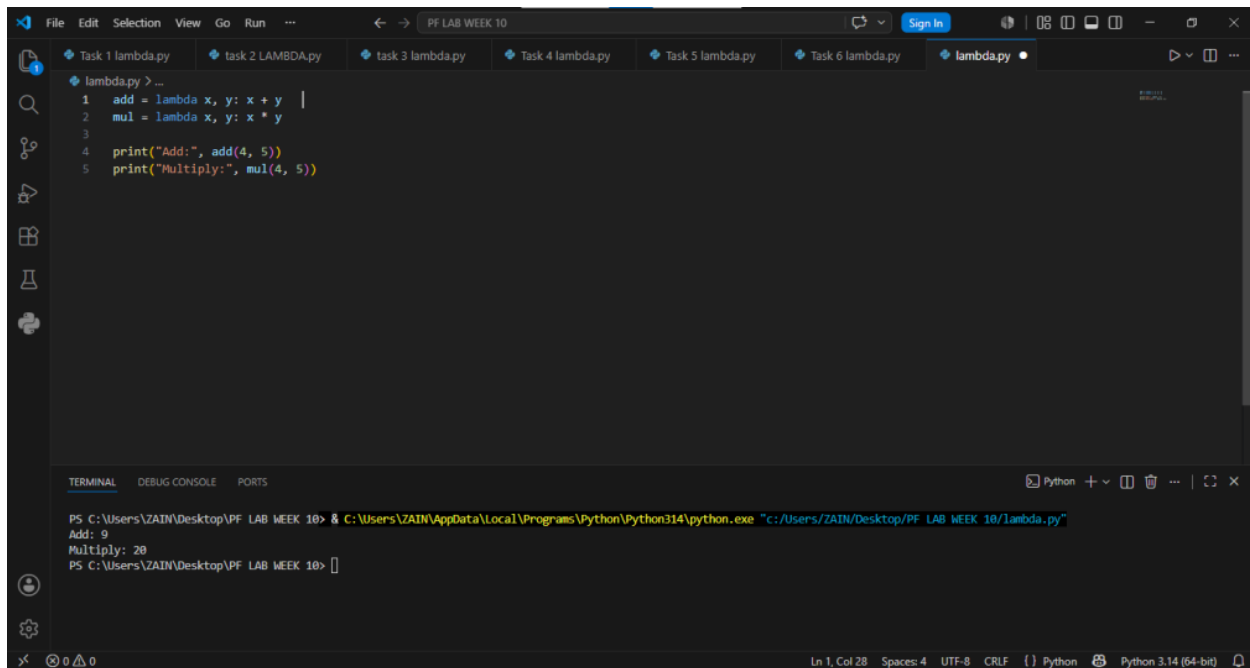
The screenshot shows a Python IDE with a file named 'Task 6 lambda.py'. The code defines a list 'a' and uses a lambda function with 'map' to square each element. The terminal output shows the command to run the script and the resulting output '[1, 4, 9, 16]'.

```
1 a = [1,2,3,4]
2 a=list(map(lambda x:x*x,a))
3 print(a)
```

```
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10> & C:\Users\ZAIN\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/ZAIN/Desktop/PF LAB WEEK 10/Task 6 lambda.py"
[1, 4, 9, 16]
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10>
```

Write a Python program using two lambda functions one to add two numbers and one to multiply two numbers

Result:



The screenshot shows a Python IDE with a file named 'lambda.py' open. The code defines two lambda functions: 'add' for addition and 'mul' for multiplication. It then prints the results of these functions applied to the numbers 4 and 5. The terminal output shows 'Add: 9' and 'Multiply: 20'.

```
1 add = lambda x, y: x + y
2 mul = lambda x, y: x * y
3
4 print("Add:", add(4, 5))
5 print("Multiply:", mul(4, 5))
```

```
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10> & C:\Users\ZAIN\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/ZAIN/Desktop/PF LAB WEEK 10/lambda.py"
Add: 9
Multiply: 20
PS C:\Users\ZAIN\Desktop\PF LAB WEEK 10>
```

Conclusion:

This lab provided practical knowledge of lambda functions and their role in simplifying Python code. We learned how to create anonymous functions for performing simple operations without defining a complete function. We also understood how lambda functions improve code readability and are commonly used with built-in functions such as `map()`, `filter()`, and `sorted()` to perform operations more efficiently.